MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC TEST – 03 (PROBABILITY)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 14: Probability** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 2 VSAQ questions (2 marks each). Section C has 2 SA questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (5 marks). Section E has 1 Applied Case Study (4 marks).
4. Show clear steps in all probability derivations and calculations.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. Two dice are rolled one after the other. The probability that the number on the first die is strictly smaller than the number on the second die is: **[1]**

Options: (a) $\frac{15}{36}$ (b) $\frac{5}{12}$ (c) $\frac{1}{2}$ (d) Both (a) and (b)

2. All kings and queens are removed from a pack of 52 cards. A card is drawn from the remaining cards. The probability of getting a black face card is: **[1]**

Options: (a) $\frac{1}{22}$ (b) $\frac{2}{44}$ (c) $\frac{1}{44}$ (d) Both (a) and (b)

3. A jar contains 54 marbles each of which is blue, green, or white. The probability of selecting a blue marble is $\frac{1}{3}$ and that of selecting a green marble is $\frac{4}{9}$. How many white marbles does the jar contain? **[1]**

Options: (a) 12 (b) 18 (c) 24 (d) 6

4. Cards numbered from 6 to 50 are put in a box. The probability of drawing a card with a prime number less than 20 is: **[1]**

Options: (a) $\frac{4}{45}$ (b) $\frac{5}{45}$ (c) $\frac{1}{9}$ (d) Both (a) and (c)

5. **Assertion (A):** The probability that a leap year has 53 Sundays is $\frac{2}{7}$.
Reason (R): A leap year has 366 days, which consists of 52 complete weeks and 2 extra days that can be any of the 7 consecutive day pairs. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

6. A card is drawn from a well-shuffled pack of 52 cards. Find the probability of getting: (i) a non-face card (ii) a red ten. **[2]**

7. It is given that in a group of 3 students, the probability of 2 students not having the same birthday is 0.992. What is the probability that the 2 students have the same birthday? **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

8. A die has its six faces marked $0, 1, 1, 1, 6, 6$. Two such dice are thrown together and the total score is recorded. (i) List all the possible distinct values of the total score. (ii) Find the probability of getting a total of 7. (iii) Find the probability of getting a total score of at least 2. **[3]**

9. A carton consists of 100 shirts of which 88 are good, 8 have minor defects, and 4 have major defects. Jimmy, a trader, will only accept shirts which are good, but Sujatha, another trader, will only reject shirts which have major defects. One shirt is drawn at random from the carton. What is the probability that: (i) it is acceptable to Jimmy? (ii) it is acceptable to Sujatha? (iii) it has major defects? **[3]**

SECTION D: LONG ANSWER / HOTS (5 Marks Each) [5 Marks]

10. A box contains 90 discs numbered from 1 to 90. If one disc is drawn at random from the box, calculate the probability that it bears: (i) a two-digit number (ii) a perfect square number (iii) a prime number less than 30 (iv) a number divisible by both 3 and 5 (v) a number divisible by 2 or 3. **[5]**

SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 11: Case Study Analysis

A social science survey recorded the ages (in completed years) of 12 senior secondary students selected for a youth council:

$$14,\ 17,\ 15,\ 21,\ 17,\ 20,\ 16,\ 18,\ 17,\ 16,\ 19,\ 20$$

One student is selected at random from this council.

Based on the above scenario, answer the following:

(i) What is the probability that the selected student is of age 18 or older (eligible to vote)? **[1]**

(ii) What is the probability that the selected student is strictly younger than 16 years? **[1]**

(iii) What is the mode of this age distribution, and what is the probability that a randomly chosen student has the modal age? **[1]**

(iv) What is the probability that the student's age is a prime number? **[1]**

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